Hannah Galocy

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Software engineer with 5+ years of experience building desktop applications, modernizing C++ and Python systems, and improving data processing performance. Skilled in debugging complex systems and improving software maintainability. Builds personal projects combining AI APIs, persistent memory, and interactive interfaces.

Work Experience

Tactical Engineering and Analysis (TEA)

Software Engineer ***Jan 2022 - Oct 2026***

- Responsible for building a distributable, modernized version of a deprecated application using Electron.js and C++.
- Led modernization efforts by rewriting legacy Python/SWIG/C++ components into clean, maintainable executables, improving performance by 20% and reducing dependencies.
- Acted proactively on unprioritized technical debt, identifying and fixing long standing bugs in recording systems, HMI interface, and Link 22 software.
- Moved TTNT networking functionality from the user interface layer into a core module and added manual and automated tests using SNMPv3.
- Guided new teammates, reviewed pull requests and provided detailed feedback when needed.
- Acquired Secret Security Clearance for position.
- Government contractor for Naval Information Warfare Center (NIWC).

San Diego State University Research Foundation (SDSURF)

Software Engineer Intern ***Nov 2020 - Dec 2021***

- Supported government software efforts as part of a classified Naval Information Warfare Center (NIWC) contract.
- Assisted in debugging and testing of custom C++ applications used in secure environments.
- Collaborated with multiple software teams across the project.

NVIDIA

Performance Engineer Intern ***Summer of 2018***

- Developed scripts for NVIDIA's performance lab to automate various benchmarks on handheld devices. This included GPU and CPU benchmarks to aid in the testing of different mobile devices.
- Used Perforce to manage Perl scripts for use across NVIDIA's performance testing teams.
- Compiled benchmark results requested by engineering teams across the company.

Projects

Navi AI Companion | Python, OpenAI API, SQLite, Flask, Piper TTS

- Developing an AI companion with LLM powered conversation, persistent SQLite memory, local speech synthesis, and a responsive animated face built with Pyglet and ModernGL.

Pancake Test Fixture | Electron, Node.js, Arduino, JSON | Senior design

- Built a desktop GUI application that exchanged JSON commands and measurements with an Arduino over USB serial, evaluated amplifier test results, and graphed data for fault diagnosis.

Skills

Languages and Frameworks:

- Python, R, C, C++, C#, Java, ARM/MIPS Assembly, Verilog, Perl, Bash, JavaScript, HTML, CSS, Electron, Node.js, React, Flask, OpenAI API, SQLite, Faster-Whisper, Piper TTS

Tools and practices:

- Git, CMake, Jenkins, GoogleTest, Automated testing, Code review, Agile, Confluence, Microsoft Office, Bitbucket, Jira, GitLab, Perforce, Raspberry Pi, Microcontrollers (AVR), FPGAs, Linux, Windows, and macOS

Education

San Diego State University ***Graduation Fall 2021***

Major: Computer Engineering | Minor: Computer Science